



RUDRAEDGE AEROSPACE

Project-RudraEdgeRE-T1(Spectra)
Daily Engineering Log Template

1. TEAM MEMBER INFORMATION

Engineer-Name:PiyushSharma

Designation/Role:RF&DSP Engineer

Task Order Reference: REA/2026/RE-T1/TRAIN-001

2. DAILY LOG ENTRY FOR

- **Date:** 05 October 2026
- **1. Assigned Task:** Electromagnetic Layer Study (RE-T1 electromagnetic-awareness architecture study)
- **2. What Was Studied:** Studied the RE-T1 electromagnetic-awareness architecture, RF signal fundamentals, frequency, wavelength, amplitude, phase, bandwidth, sampling, I/Q representation, spectrum analysis, and the conceptual relationship between the RF and DSP processing layers.
- **3. What Was Implemented:** A high-level conceptual RF architecture for the RE-T1 electromagnetic-awareness layer was prepared. The architecture defines the conceptual flow from the electromagnetic environment to RF information, signal representation, digital signal data, and the DSP processing layer. The RF-to-DSP interface was also documented as part of the architecture study.
- **4. Tools / Libraries Use:**VS-Code, Git, GitHub. Markdown.
- **5. Output File Name:** day01_em_layer/rf_architecture_notes.md
Path- https://github.com/piyush66oo1000-hub/RE-T1_daily_task
- **6. Problems Encountered:** None.
- **7. How Problem Was Resolved:** No problem encountered; the task was completed successfully.

- **8. Next Task Scheduled:** 06 October 2026 — RF Fundamentals & Link Metrics (Frequency, bandwidth, sampling, noise and SNR calculations).
- **9. Personal Expenditure:** ₹0 (Verified Mandate)

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastru

Candidate/Engineer

Candidate/Engineer Signature: 
Full Name: Piyush Sharma

Date: 05/10/2026

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

- ☐ Approved / Completed
- ☐ Needs Revision / Delayed
- ☐ Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____ / ____ / 2026